Introduction to Descriptive Statistics

Learning Objectives

By the end of this session, you will be able to:

- Understand **measures of descriptive statistics**: Mean, Median, Mode, Standard Deviation, etc.
- Learn how to **derive and interpret** these measures.
- Use **Python** for hands-on calculation of statistics.

Multiple Choice Questions (MCQs)

What is the primary goal of descriptive statistics?

A) To make inferences about a population

B) To describe and summarize data

C) To test hypotheses

D) To establish causality

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Answer: B) To describe and summarize data
Descriptive statistics aims to provide a summary of the main features of a dataset, making it easier to understand.

What is Descriptive Statistics?

Descriptive statistics summarize and describe the basic features of a dataset.

Purpose:

- Simplifies complex datasets.
- Identifies patterns, trends, and outliers.
- Forms the foundation for more advanced statistical modeling.

Types of Descriptive Statistics

Measures of Central Tendency: Mean, Median, Mode.

Measures of Dispersion: Range, Variance, Standard Deviation.

Mean

- **Formula:**

$$\text{Mean}(\mu) = \frac{\sum_{i=1}^{n} x_i}{n}$$

Where x_i are the data points and n is the total number of observations.

- **Interpretation:**

Represents the **average** value of a dataset.

$$\text{Mean}(\mu) = \frac{\sum_{i=1}^{n} x_i}{n}$$

Component Breakdown

Symbol	Name	Meaning
$\text{Mean}(\mu)$	**Mean** or **Population Mean**	This is the value you are calculating. The Greek letter μ **(mu)** specifically denotes the **population mean** (the average of *all* possible values). If you were calculating the mean of a **sample** (a subset of the population), the symbol would typically be $\bar{x}$ **(x-bar)**.
$\sum$	**Capital Sigma** (Summation)	This is the **summation** operator. It means you must **add up** everything that follows it.
$i = 1$	**Index Start**	This indicates that the summation starts with the **first** data point in the set (where the index i is 1).
n (above $\sum$)	**Index End**	This indicates that the summation stops at the **last** data point, meaning you sum up to the n-th data point.
x_i	**Data Point**	This represents the individual values in your data set. x_1 is the first value, x_2 is the second, and so on, up to x_n.
$\sum_{i=1}^{n} x_i$	**The Numerator**	This entire expression means: **"Sum up all the data points from the first (x_1) to the last (x_n)."** In simple terms, it's the **total sum** of all the numbers in your set.
n (in the denominator)	**Count**	This is the **total number of data points** in the set.

1. Test Scores

Element (x_i)	Value
x_1	**85**
x_2	**92**
x_3	**78**
x_4	**95**
x_5	**80**

Data Set: $\{85, 92, 78, 95, 80\}$

Step 1: Sum all the values ($\sum x_i$)

We add up all the individual test scores (x_i):

$$\text{Sum} = 85 + 92 + 78 + 95 + 80$$

$$\text{Sum} = 430$$

Step 2: Count the number of values (n)

There are five test scores in the data set:

$$n = 5$$

Step 3: Divide the sum by the count

We substitute the values into the mean formula:

$$\bar{x} = \frac{430}{5}$$

$$\bar{x} = 86.0$$

The arithmetic mean (average) of the test scores is **86.0**.

2. Daily Temperatures (°C)

Element (x_i)	Value
x_1	20
x_2	25
x_3	22
x_4	27
x_5	26

3. Number of Books Read

Element (x_i)	Value
x_1	3
x_2	1
x_3	4
x_4	2
x_5	5

4. Customer Wait Times (Minutes)

Element (x_i)	Value
x_1	4.5
x_2	6.0
x_3	3.5
x_4	5.0
x_5	4.0

5. Number of Cars Sold per Day

Element (x_i)	Value
x_1	10
x_2	10
x_3	10
x_4	10
x_5	10

Median

Definition: The **middle value** when data is sorted in ascending order.

When to Use: Best for **skewed** data or when there are **outliers**.

Example:

*For data [5, 10, 15, 20, 100], the **median** is 15, while the **mean** is skewed by 100.*

1. Test Scores (Odd Count)

Element (x_i)	Value
x_1	85
x_2	92
x_3	78
x_4	95
x_5	80
x_6 (Outlier)	**10**
x_7	88

Data Set: $\{85, 92, 78, 95, 80, \mathbf{10}, 88\}$

Total Count (n): 7

2. Daily Temperatures (°C) (Even Count)

Element (x_i)	Value
x_1	20
x_2	25
x_3	22
x_4	27
x_5	26
x_6	24
x_7 (Outlier)	**45**
x_8	23

Data Set: $\{20, 25, 22, 27, 26, 24, \mathbf{45}, 23\}$

Total Count (n): 8

3. Number of Books Read (Odd Count with a high outlier)

Element (x_i)	Value
x_1	3
x_2	1
x_3	4
x_4	2
x_5	5
x_6	3
x_7 (Outlier)	**30**

Data Set: $\{3, 1, 4, 2, 5, 3, \mathbf{30}\}$

Total Count (n): 7

Mode

Definition: The value that appears **most frequently** in a dataset.

When to Use: Useful for **categorical data** or identifying the most common occurrence.

Example:
If the most common loan amount is $10,000, that's the mode.

Mode Calculation Results

Sample Data Set	Data Set	Frequency Count	Mode
1. Unimodal	$\{5, 8, \mathbf{12}, 5, \mathbf{12}, 15, \mathbf{12}\}$	**12** appears 3 times.	**12**
2. Bimodal	$\{\mathbf{4}, 7, \mathbf{9}, 10, \mathbf{4}, 7, \mathbf{9}, 11\}$	**4** appears 2 times; **9** appears 2 times.	**4 and 9**
3. No Mode	$\{21, 25, 28, 22, 29, 26\}$	All numbers appear 1 time.	**No Mode**
4. Decimals/Currency	$\{1.50, \mathbf{2.25}, 1.50, \mathbf{2.25}, 3.00\}$	**1.50** appears 2 times; **2.25** appears 2 times.	**1.50 and 2.25**
5. Multimodal (w/ Outliers)	$\{\mathbf{100}, 1, \mathbf{100}, \mathbf{200}, 2, \mathbf{200}, \mathbf{100}\}$	**100** appears 3 times; **200** appears 2 times.	**100**

Standard Deviation

Standard deviation is a statistical measure of data dispersion that quantifies how much individual data points differ from the mean (average) of a dataset.

A low standard deviation signifies that data points are clustered closely around the mean, **indicating consistency,** while a **high standard deviation** shows that the data points are spread out over a wider range, **indicating greater variability.**

It is calculated as the square root of the **variance**, which is the average of the squared differences from the mean.

Variance

- **Definition:** The **average of squared differences** from the mean.
- **Formula:**

$$\text{Variance}(\sigma^2) = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{n}$$

Range and Interquartile Range (IQR)

- **Range:**

$$\text{Range} = \text{Max} - \text{Min}$$

Shows the **spread** between the smallest and largest values.

- **IQR (Interquartile Range):**

Measures the spread of the **middle 50%** of data.

$$\text{IQR} = Q3 - Q1$$

1. Range,
2. IQR (**Interquartile Range**),
3. Variance, and
4. Standard Deviation (SD)

are the four main ways to measure **dispersion** (or spread) in a data set.

Example Data Set: Daily Sales (Units Sold)

We will use the following set of 7 daily sales figures for a small shop.

$$\text{Data Set}(X) = \{12, 18, 20, 25, 25, 30, 42\}$$

Total number of observations (n) = 7.

1. Range

The **Range** is the simplest measure of dispersion. It's the difference between the highest and lowest values in the data set.

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

- **Maximum Value:** 42
- **Minimum Value:** 12

$$\text{Range} = 42 - 12 = \mathbf{30}$$

The sales figures span a range of 30 units.

Example Data Set: Daily Sales (Units Sold)

We will use the following set of 7 daily sales figures for a small shop.

$$\text{Data Set}(X) = \{12, 18, 20, 25, 25, 30, 42\}$$

Total number of observations (n) = 7.

2. Interquartile Range (IQR)

The **IQR** measures the spread of the **middle 50%** of the data. It is less affected by outliers than the range.

$$\text{IQR} = Q_3 - Q_1$$

Step 2a: Find the Quartiles (Q_1 and Q_3)

The data must first be sorted, which it already is: $\{12, 18, 20, 25, 25, 30, 42\}$.

1. **Find the Median (Q_2):** Since $n = 7$ (odd), the median is the middle value, which is the $\frac{7+1}{2} = 4^{\text{th}}$ position.

$$\{12, 18, 20, \mathbf{25}, 25, 30, 42\} \implies \mathbf{Q_2 = 25}$$

2. **Find the First Quartile (Q_1):** This is the median of the lower half of the data (the values before Q_2): $\{12, 18, 20\}$.

$$\{12, \mathbf{18}, 20\} \implies \mathbf{Q_1 = 18}$$

3. **Find the Third Quartile (Q_3):** This is the median of the upper half of the data (the values after Q_2): $\{25, 30, 42\}$.

$$\{25, \mathbf{30}, 42\} \implies \mathbf{Q_3 = 30}$$

Step 2b: Calculate IQR

$$\text{IQR} = Q_3 - Q_1 = 30 - 18 = \mathbf{12}$$

The middle 50% of daily sales figures has a spread of 12 units.

Example Data Set: Daily Sales (Units Sold)

We will use the following set of 7 daily sales figures for a small shop.

$$\text{Data Set}(X) = \{12, 18, 20, 25, 25, 30, 42\}$$

Total number of observations (n) = 7.

3. Variance (s^2) and 4. Standard Deviation (s)

Variance and Standard Deviation measure the average distance of each value from the mean. We will calculate the **Sample Variance (s^2)** and **Sample Standard Deviation (s)** using the $n-1$ denominator (Degrees of Freedom).

Step 3a: Calculate the Mean ($\bar{x}$)

$$\bar{x} = \frac{\sum x}{n} = \frac{12 + 18 + 20 + 25 + 25 + 30 + 42}{7} = \frac{172}{7} \approx \mathbf{24.57}$$

Step 3b: Calculate Deviations and Squared Deviations

x_i (Value)	$x_i - \bar{x}$ (Deviation)	$(x_i - \bar{x})^2$ (Squared Deviation)
12	$12 - 24.57 = -12.57$	158.00
18	$18 - 24.57 = -6.57$	43.16
20	$20 - 24.57 = -4.57$	20.88
25	$25 - 24.57 = 0.43$	0.18
25	$25 - 24.57 = 0.43$	0.18
30	$30 - 24.57 = 5.43$	29.48
42	$42 - 24.57 = 17.43$	303.80
Sum ($\sum$)	$\approx \mathbf{0}$	$\sum(x_i - \bar{x})^2 \approx 555.68$

Step 3c: Calculate the Sample Variance (s^2)

$$\text{Variance } (s^2) = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$$\text{Variance } (s^2) = \frac{555.68}{7-1} = \frac{555.68}{6} \approx \mathbf{92.61}$$

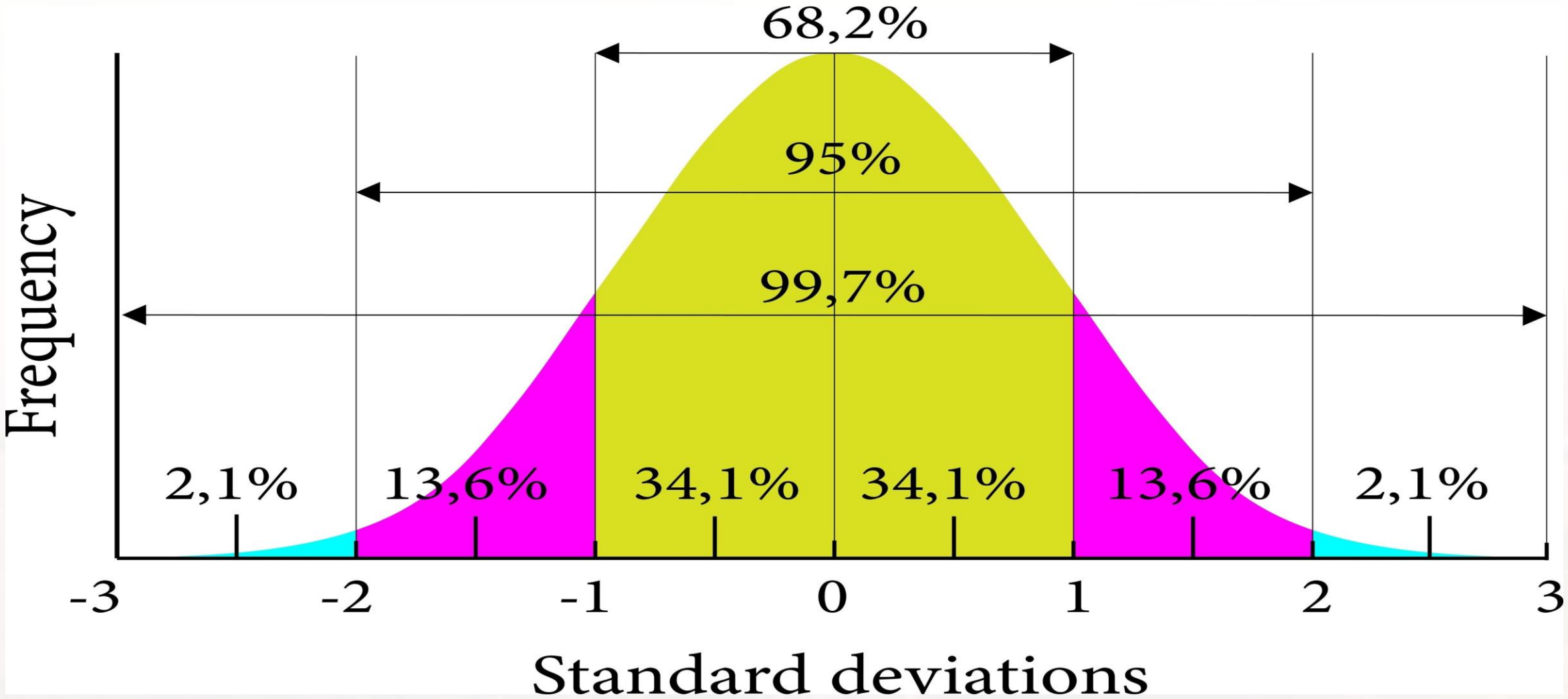
Step 4: Calculate the Sample Standard Deviation (s)

The Standard Deviation is the square root of the Variance.

$$\text{Standard Deviation } (s) = \sqrt{\text{Variance}} = \sqrt{s^2}$$

$$\text{Standard Deviation } (s) = \sqrt{92.61} \approx \mathbf{9.62}$$

This means that, on average, a day's sales deviates from the mean sales figure ($\bar{x} = 24.57$) by about 9.62 units.


68,2%
95%
99,7%
Frequency
2,1%
13,6%
34,1%
34,1%
13,6%
2,1%
-3
-2
-1
0
1
2
3
Standard deviations

Measure of Dispersion	Result	Interpretation
Range	30	The entire data set spans 30 units (from 12 to 42).
IQR	12	The middle 50% of sales are spread over 12 units.
Variance (s^2)	92.61	The average of the squared differences from the mean (units squared, harder to interpret directly).
Standard Deviation (s)	**9.62**	The average distance each sales figure is from the mean ($\bar{x}=24.57$) in the original units.

Questions?

1. Which measure of central tendency is most affected by outliers?

A) Median B) Mode C) Mean D) Range

2. What does a standard deviation of 0 indicate?

A) All values are different B) Data is widely spread C) All values are the same D) There are outliers in the data

3. If the mean and median are equal, the data is likely:

A) Skewed right B) Symmetrical C) Skewed left D) Uniform

Questions?

1. Which measure of central tendency is most affected by outliers?

A) Median B) Mode C) Mean ✅ D) Range

2. What does a standard deviation of 0 indicate?

A) All values are different B) Data is widely spread C) All values are the same ✅ D) There are outliers in the data

3. If the mean and median are equal, the data is likely:

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Python Implementation

- Credit Risk Analysis for LendingClub.com
- You are a data scientist working at LendingClub.com and you have been asked to perform descriptive statistics on the data….

Sl.no	Variable Name	Description
1	credit.policy	1 if the customer meets the credit underwriting criteria of LendingClub.com, 0 otherwise.
2	purpose	The purpose of the loan (e.g., "credit_card", "debt_consolidation", "educational", etc.).
3	int.rate	The interest rate of the loan as a proportion (e.g., 11% is stored as 0.11). Higher risk equals higher rates.
4	installment	The monthly installments ($) owed by the borrower if the loan is funded.
5	log.annual.inc	The natural log of the self-reported annual income of the borrower.
6	dti	The debt-to-income ratio of the borrower (debt amount divided by annual income).
7	fico	The FICO credit score of the borrower.
8	days.with.cr.line	The number of days the borrower has had a credit line.
9	revol.bal	The borrower's revolving balance (amount unpaid at the end of the credit card billing cycle).
10	revol.util	The borrower's revolving line utilization rate (credit line used relative to total credit available).
11	inq.last.6mths	The borrower's number of inquiries by creditors in the last 6 months.
12	delinq.2yrs	The number of times the borrower had been 30+ days past due on a payment in the past 2 years.
13	pub.rec	The borrower's number of derogatory public records (bankruptcy filings, tax liens, judgments).
14	not.fully.paid	Binary variable indicating if the loan was not paid back in full (defaulted or charged off).

Session Summary

In this session, we have discussed about:

- **Descriptive statistics** help summarize and interpret datasets.
- **Central tendency** measures (Mean, Median, Mode) describe data location.
- **Dispersion** measures (Standard Deviation, Variance, IQR) show variability.

Reference Links

- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- Top 20 Frequently Asked Questions on Descriptive Statistics for Data Science Aspirants.
- How Much Statistics Do You Really Need for Data Science?
- Phases of data science and analytics.